WEEK 2 · DAY 5 · OVERVIEW

Case Study

Credit Card Default



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ABOUT 50 MIN WITH EXERCISES / 12 SECTIONS / 4 EXERCISES

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Problem Statement

BUSINESS CONTEXT

Days 6 to 9 built the pieces: odds, the confusion matrix, a threshold chosen for a recall target, a pruned tree.

Today the desk runs the whole job once, as a write-up would.

Which model ranks the accounts that defaulted highest, how many accounts must it flag to catch 70 percent of them, and what does it miss?

The accounts are real, from Taiwan, April to September 2005, so they describe consumer credit there and then, not US credit today.

SEX, **EDUCATION**, **MARRIAGE** and **AGE** are never features: a model using them would sort people by them, and leaving them out does not make a model neutral, because other columns can carry the same information.

A flag means "flagged at threshold t", never a credit decision about anyone.

OBJECTIVES

Tame the heavy-tailed amounts, fit a logistic regression and two trees, and choose on validation accounts a threshold that catches 70 percent of the defaults.

Pick one model, open the test accounts once, describe its flags by the **SEX** label, and say where it fails.

ABOUT 50 MIN WITH EXERCISES / 12 SECTIONS / 4 EXERCISES

Your Route Through Day 5

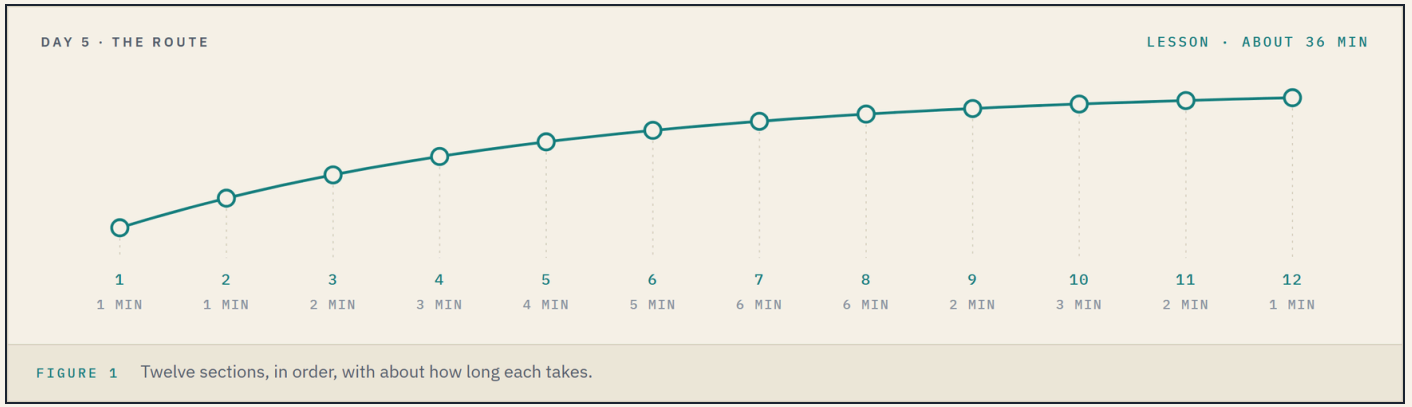


FIGURE 1 Twelve sections, in order, with about how long each takes.

	SECTION	ABOUT
1	Setting Up	1 min
2	Loading the Data	1 min
3	Data Overview	2 min
4	Exploratory Data Analysis	3 min
5	Data Preparation for Modeling	4 min
6	Model Building	5 min

	SECTION	ABOUT
7	Model Performance Improvement	6 min
8	Final Model Selection	6 min
9	The Test Rows, Opened Once	2 min
10	The Person Columns	3 min
11	Where the Model Fails	2 min
12	Insights and Conclusion	1 min

Time. About 36 minutes for the lesson, then about 12 for the four exercises.

Fix the seed and load the libraries.

Setting Up

THE SEED AND THE DATA'S LOCATION

```
import os
from pathlib import Path
os.environ["OMP_NUM_THREADS"] = "1"
os.environ["OPENBLAS_NUM_THREADS"] = "1"
SEED = 42
DATA = "../../../data/"
if not Path(DATA + "benchmark.csv").exists():
    DATA = "https://raw.githubusercontent.com/lenamonj/python-fixed-income/main/data/"
```

THE LIBRARIES THE CASE STUDY LEANS ON

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler, QuantileTransformer
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
```

Loading the Data

READ THE CARD FILE AND CHOOSE THE FEATURES

```
card = pd.read_csv(DATA + "credit_card_default.csv")
FEATURES = [column for column in card.columns if column not in ["ID", TARGET] + PERSON_COLUMNS]
print(len(FEATURES), "features")
```

```
19 features
```

THE DATA DICTIONARY

- **LIMIT_BAL**: credit limit, NT dollars
- **PAY_0**, **PAY_2** to **PAY_6**: repayment status, September back to April 2005, where -1 is paid on time, 1 to 9 is months late, and -2 and 0 are undocumented (used as numbers, a modelling choice, as on Day 6)
- **BILL_AMT1** to **BILL_AMT6**: the bill, NT dollars, September back to April
- **PAY_AMT1** to **PAY_AMT6**: the amount paid, NT dollars, September back to April
- **default payment next month**: 1 if the account defaulted the following month, **the target**
- **ID**, **SEX**, **EDUCATION**, **MARRIAGE**, **AGE**: never features

Data Overview

SHAPE OF THE FILE

```
print(card.shape)
```

```
(30000, 25)
```

WHAT THE CHECK FOUND

Nothing is missing, and the 35 repeated rows are different accounts under their own IDs, kept as on Day 9.

6,636 accounts defaulted, 22.12 percent.

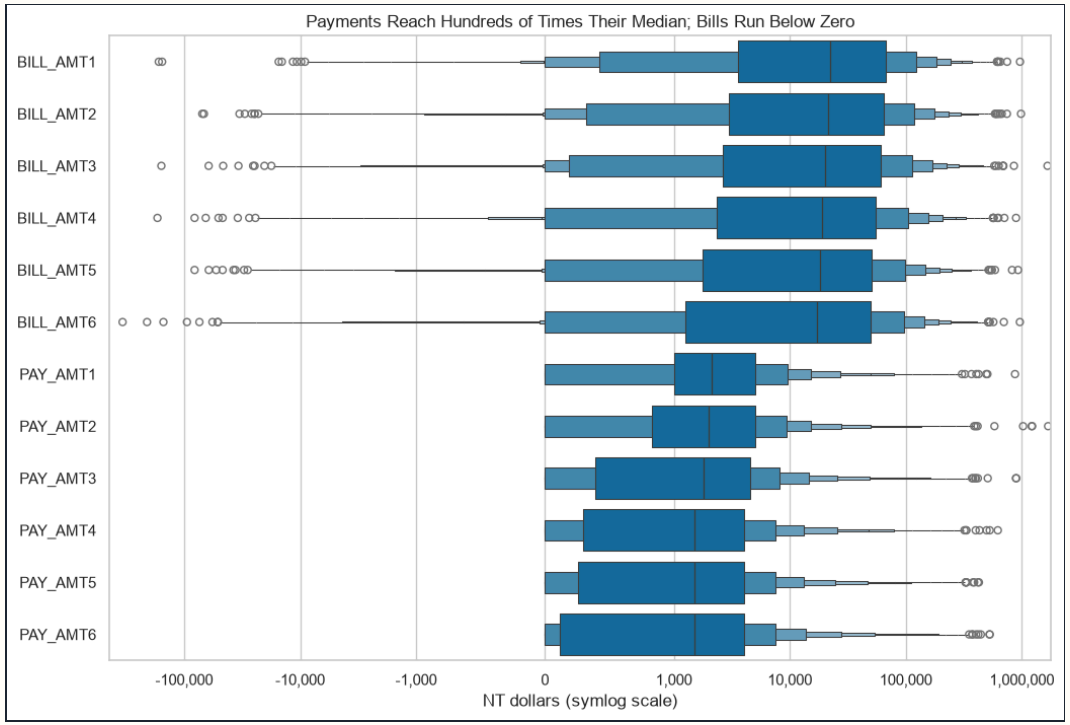
THE THREE AMOUNTS, SUMMARISED

	LIMIT_BAL	BILL_AMT1	PAY_AMT1
count	30,000.0000	30,000.0000	30,000.0000
mean	167,484.3227	51,223.3309	5,663.5805
std	129,747.6616	73,635.8606	16,563.2804
min	10,000.0000	-165,580.0000	0.0000
25%	50,000.0000	3,558.7500	1,000.0000
50%	140,000.0000	22,381.5000	2,100.0000
75%	240,000.0000	67,091.0000	5,006.0000
max	1,000,000.0000	964,511.0000	873,552.0000

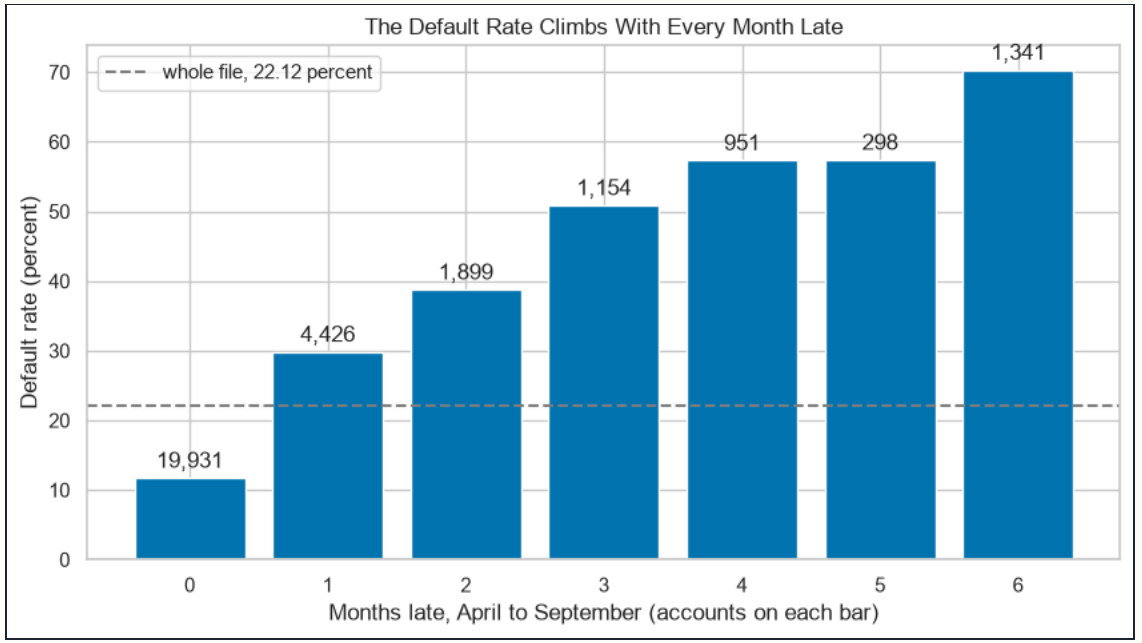
The file is clean; the amounts are not well behaved, so the exploration looks at their shape first.

Exploratory Data Analysis

TWELVE AMOUNTS ON A SYMLOG AXIS



DEFAULT RATE BY MONTHS LATE



Status separates the outcomes most, as Days 6 and 9 found; the amounts carry theirs in long tails, so the logistic model needs them tamed and the trees do not.

Data Preparation for Modeling

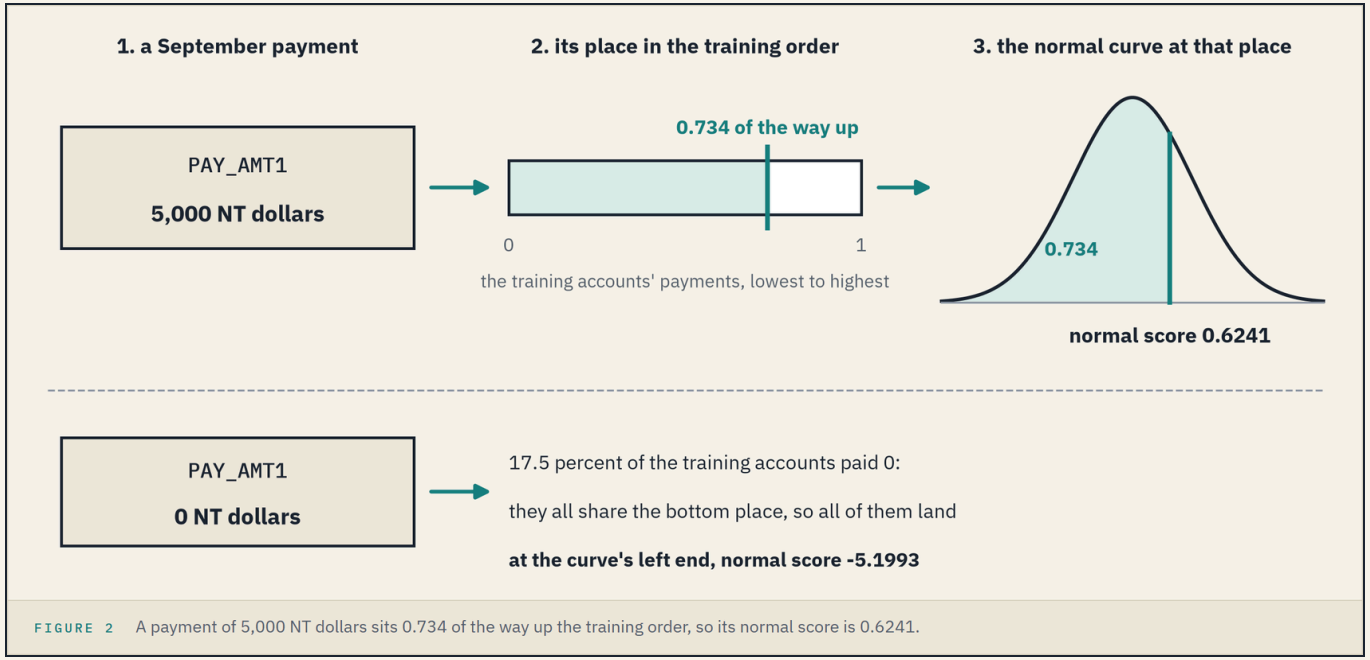
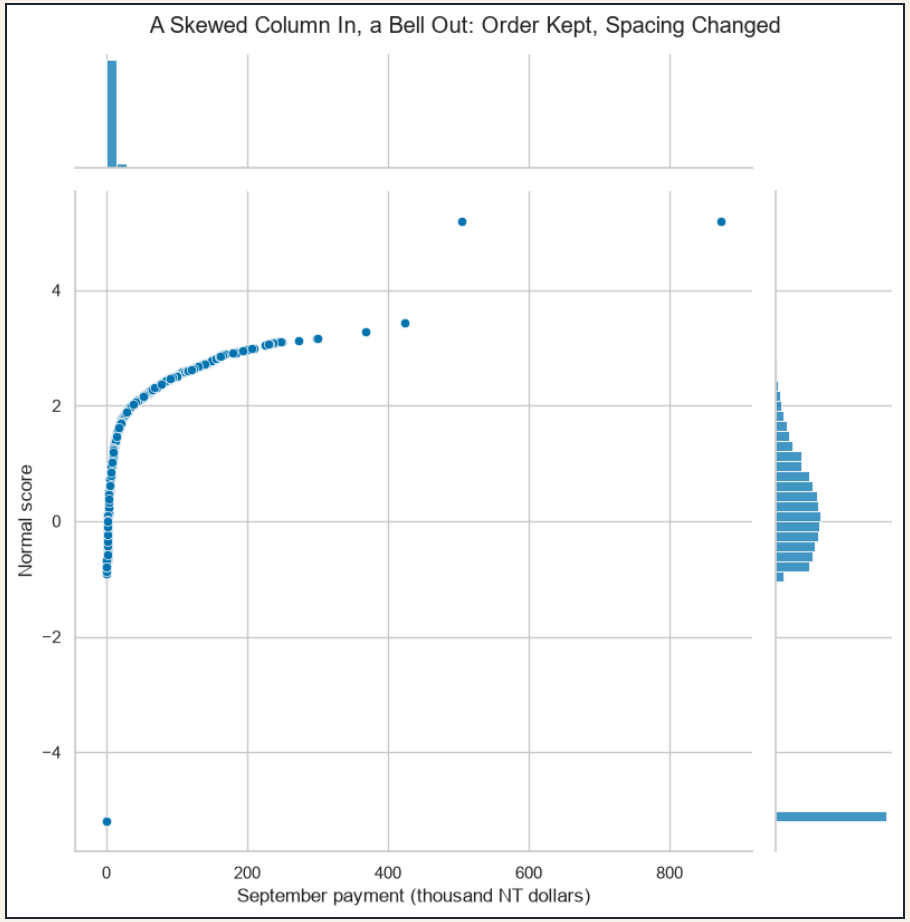
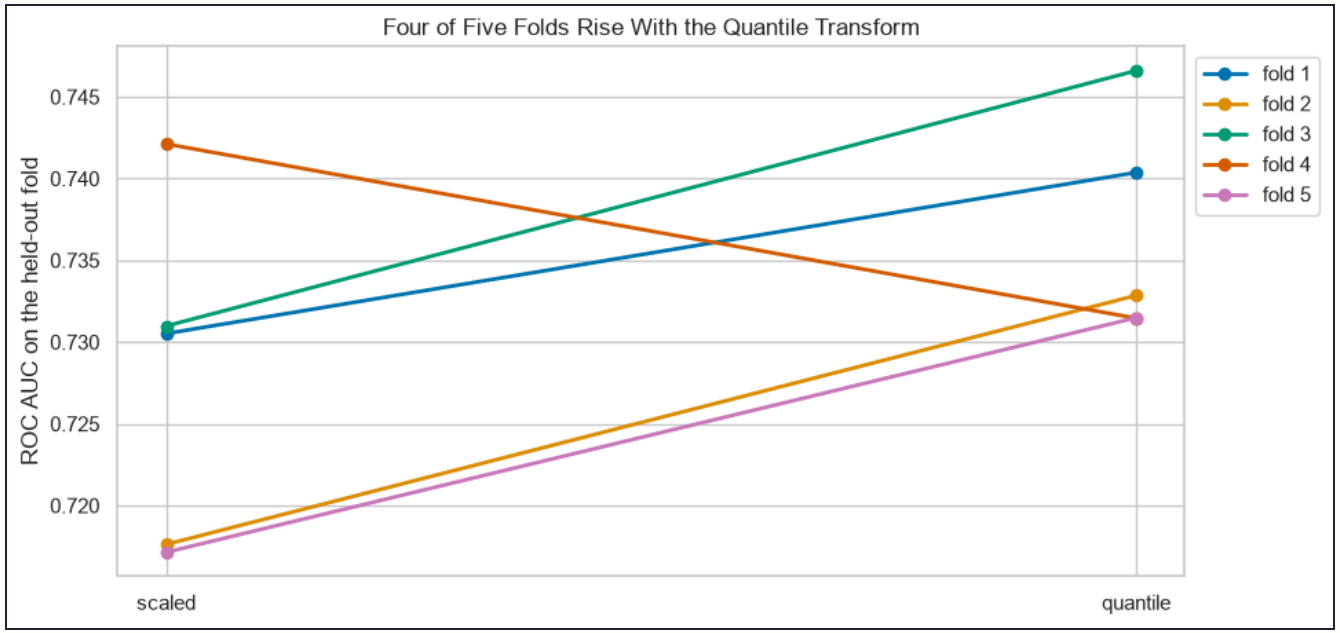


FIGURE 2 A payment of 5,000 NT dollars sits 0.734 of the way up the training order, so its normal score is 0.6241.



Model Building

EACH FOLD, SCALER AGAINST QUANTILE TRANSFORM



THE UNPRUNED TREE, TRAINED AND VALIDATED

	TRAINING	VALIDATION
ROC AUC	0.9997	0.6158
Accuracy	0.9866	0.7262
Recall	0.9997	0.4160
Precision	0.9431	0.3887
F1	0.9706	0.4019

THE VERDICT ON IT

ROC AUC 0.9997 on the training accounts and 0.6158 on validation: Day 9's full balanced tree, a memoriser again.

So the tree is pruned.

Model Performance Improvement

PRUNE BY CROSS-VALIDATION

```
alphas = [0.0002, 0.0005, 0.001, 0.002, 0.005]
search = GridSearchCV(DecisionTreeClassifier(class_weight="balanced", random_state=SEED), {"ccp_alpha": alphas},
                      cv=folds, scoring="roc_auc")
print(pd.Series(search.cv_results_["mean_test_score"], index=alphas).round(4).to_dict())
print(search.best_params_, " Depth:", pruned_tree.get_depth(), " Leaves:", pruned_tree.get_n_leaves())
```

```
{0.0002: 0.6838, 0.0005: 0.7589, 0.001: 0.7598, 0.002: 0.7496, 0.005: 0.7369}
{'ccp_alpha': 0.001}  Depth: 6  Leaves: 12
```

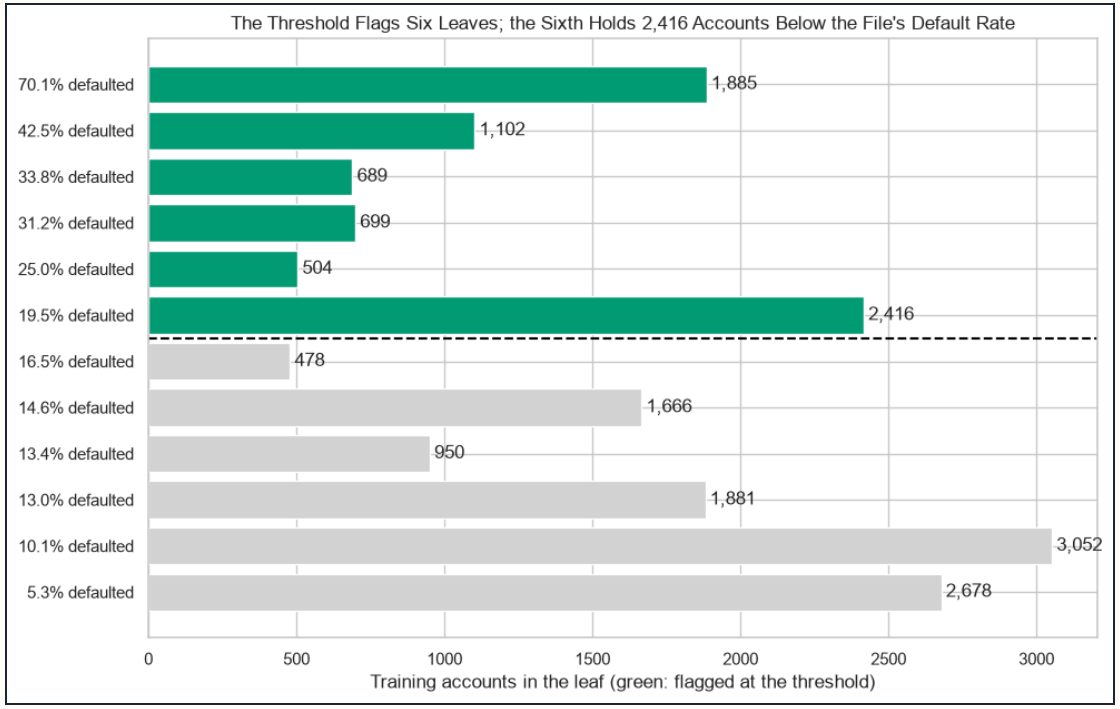
THRESHOLDS FOR A RECALL OF 0.70

	THRESHOLD	FLAGGED
scaled logistic	0.2025	2843
quantile logistic	0.2011	2506

12 leaves, 6 questions deep: Day 9's post-pruned tree, the same settings and the same tree.

Final Model Selection

TWELVE LEAVES, FLAGGED ONES IN COLOUR



THE FINALISTS ON VALIDATION ACCOUNTS

	QUANTILE LOGISTIC	DEFAULT TREE	PRUNED TREE
ROC AUC	0.7291	0.6158	0.7627
Accuracy	0.6708	0.7262	0.6945
Recall	0.7001	0.4160	0.7069
Precision	0.3707	0.3887	0.3938
F1	0.4847	0.4019	0.5058

HOW THEY COMPARE

The pruned tree leads on ROC AUC, 0.7627 against 0.7291, and at recall 0.7069 its precision is 0.3938 against the logistic's 0.3707.

The pruned tree goes forward: the highest validation ROC AUC, the fewest flags for the recall asked, and twelve rules a reader can check.

Score the chosen threshold on unseen accounts.

The Test Rows, Opened Once

OPEN THE TEST ACCOUNTS ONCE

```
test_comp = pd.concat([model_performance_classification(models[name], X_test, y_test, t).T for name, t in finalists.items()], axis=1)
test_comp.columns = list(finalists)
test_comp
```

ONE LINE PER FINALIST ACROSS THE THREE SPLITS

```
splits = {"training": (X_train, y_train), "validation": (X_valid, y_valid), "test": (X_test, y_test)}
plt.figure(figsize=(9, 7))
for name in finalists:
    auc_by_split = [roc_auc_score(target, models[name].predict_proba(rows)[:, 1]) for rows, target in splits.values()]
    plt.plot(list(splits), auc_by_split, marker="o", linewidth=2.5, color=COLOURS[name], label=name)
    plt.text(2.06, auc_by_split[2], f"{auc_by_split[2]:.4f}", va="center")
plt.ylabel("ROC AUC")
```

THE QUESTION

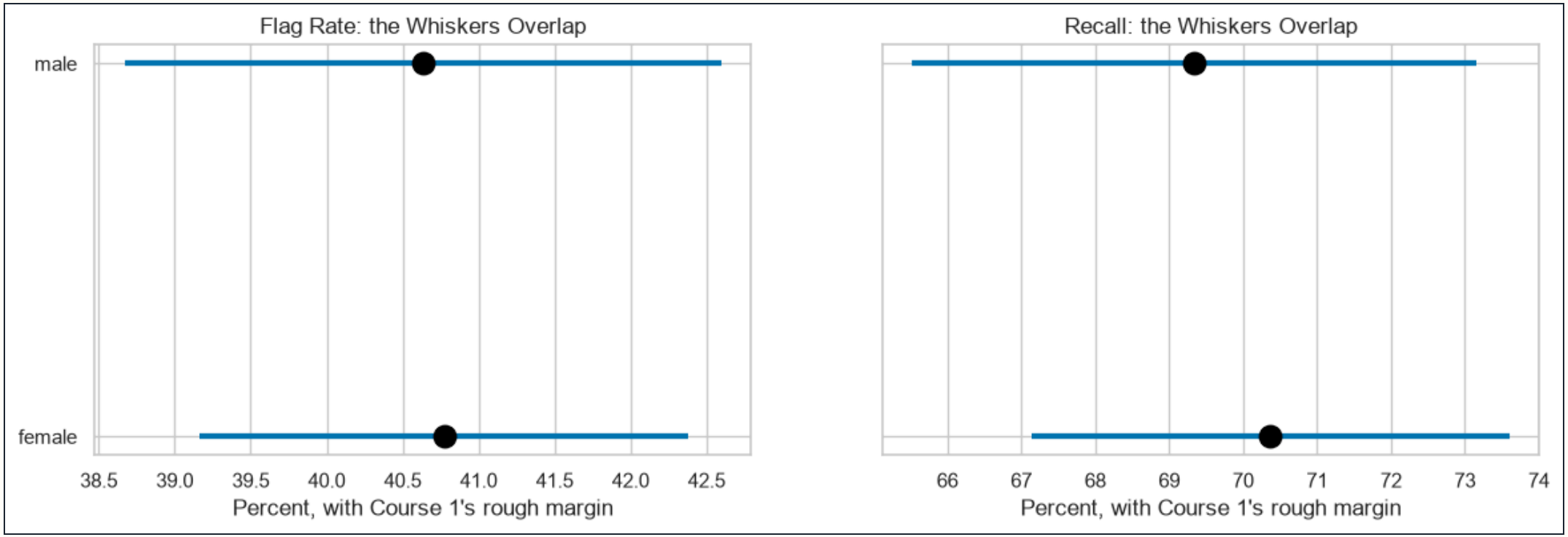
On the validation accounts the pruned tree's threshold catches 70.69 percent of the defaults.

Q. Write down its recall on the 6,000 test accounts.

Describe flags by the SEX label, with margins.

The Person Columns

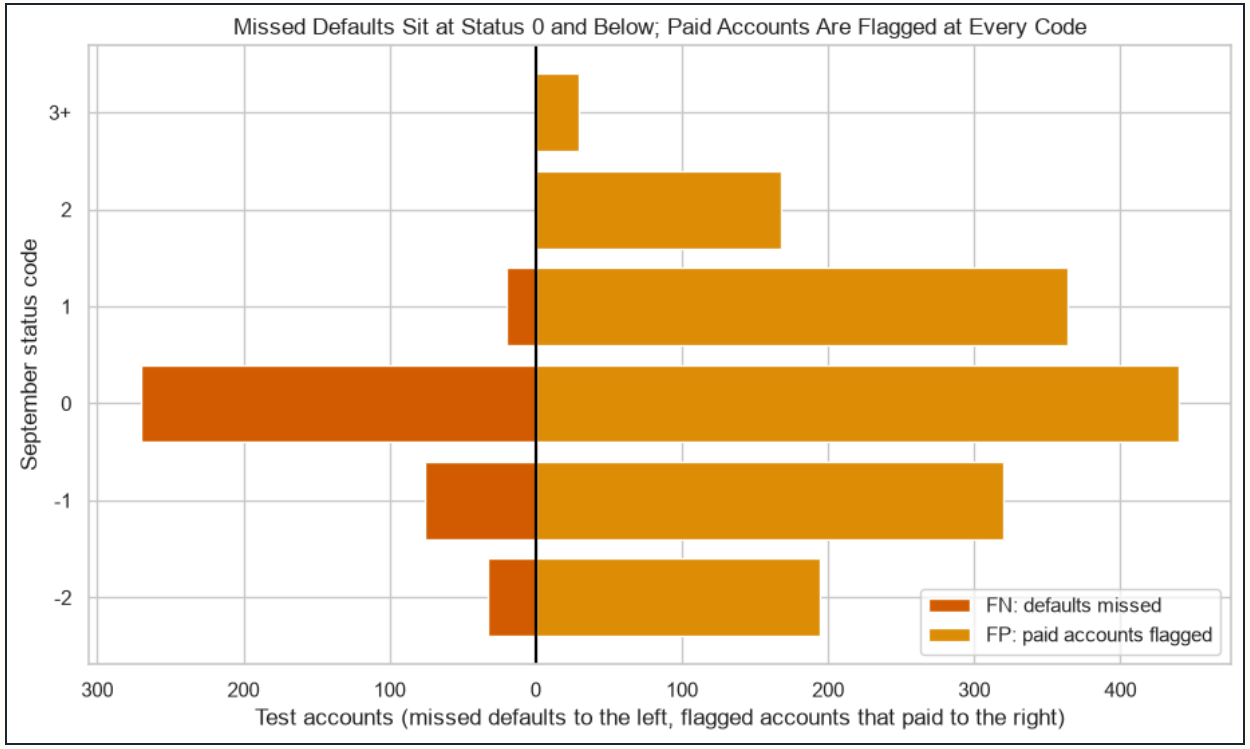
EACH LABEL'S RATES WITH THEIR MARGINS



Flag rates of 40.77 percent for the female label and 40.63 for the male, recall of 70.37 and 69.34 percent: both gaps sit well inside the margins.

Where the Model Fails

MISSED DEFAULTS AND FLAGGED PAYERS BY SEPTEMBER STATUS



COUNTS BY STATUS CODE

PAY_0	FN	FP
-2	33	194
-1	76	320
0	270	440
1	20	364
2	0	168
3	0	29

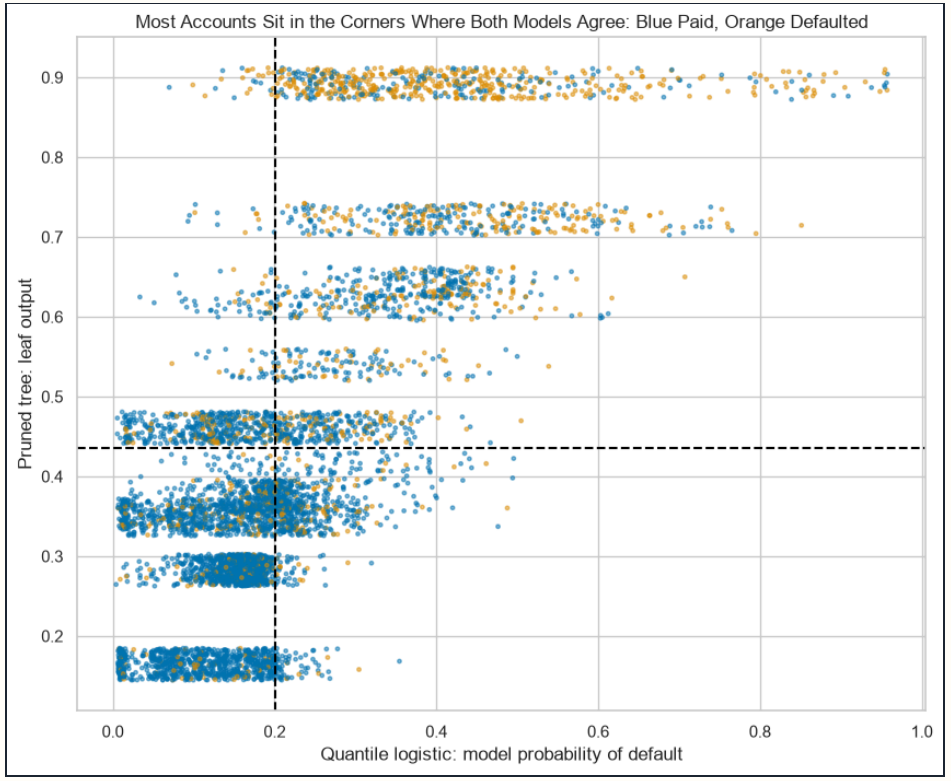
WHERE THE MISSES SIT

379 of the 399 missed defaults sit at codes 0 or below, 270 at code 0.

Every default at code 2 or more is caught, 439 of them.

Insights and Conclusion

WHERE THE TWO MODELS AGREE



ACCOUNTS FLAGGED BY BOTH, ONE, OR NEITHER

```
                      size  sum
logistic tree
False    False        2893  270
         True          601  128
True     False         725  119
         True         1781  810
Same flag: 4674  Correlation: 0.71
```

READING THE CHART

A **quadrant chart** cuts a scatter at each model's threshold.

The two models give the same flag to 4,674 of 6,000 accounts; both flag 1,781, of which 810 defaulted; neither flags 2,893, of which 270 defaulted.

A model describes the file it was fitted on, and this one says September's status first and the rest a little.

You Can Now

- Run a classification case study from the data to the test accounts, choosing everything on validation accounts.
- Tame heavy tails with **`QuantileTransformer`** inside a pipeline, and read a letter-value plot on a symlog axis.
- Choose a threshold for a recall target and compare models on a cumulative gains curve.
- Read a pruned tree's leaves with **`.apply`**, and describe a model's flags by group with a margin.
- Say where a model fails, and what its file cannot support.

NEXT
WEEK 3 · DAY 1

Dated Data and Walk-Forward Validation

Day 11 adds the calendar: dated data and validation that walks forward.